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In [1]: # # This mounts your Google Drive to the Colab VM.
# from google.colab import drive
# drive.mount('/content/drive')

# # TODO: Enter the foldername in your Drive where you have saved the unz
# # assignment folder, e.g. 'cs6353/assignments/assignment1/'
# FOLDERNAME = 'assignment1'
# assert FOLDERNAME is not None, "[!] Enter the foldername."

# # Now that we've mounted your Drive, this ensures that
# # the Python interpreter of the Colab VM can load
# # python files from within it.
# import sys
# sys.path.append('/content/drive/My Drive/{}'.format(FOLDERNAME))

# This downloads the CIFAR-10 dataset to your Drive
# if it doesn't already exist.
# %cd /content/drive/My\ Drive/$FOLDERNAME/cs6353/datasets/
# !bash get_datasets.sh
# %cd /content/drive/My\ Drive/$FOLDERNAME

# ----- Laptop -----
# %cd "/home/jared/Desktop/Fall2025/CS-6353/Homeworks/assignment1/cs6353/"
# !bash get_datasets.sh
# %cd "/home/jared/Desktop/Fall2025/CS-6353/Homeworks/assignment1"
# ----- Laptop -----

# ----- Desktop -----
%cd "/home/jared/CS-6353/Homeworks/assignment1/cs6353/datasets/"
!bash get_datasets.sh
%cd "/home/jared/CS-6353/Homeworks/assignment1/"
# ----- Desktop -----

# Install requirements from colab_requirements.txt
# TODO: Please change your path below to the colab_requirements.txt file

# ----- Desktop -----
! python -m pip install -r "/home/jared/CS-6353/Homeworks/assignment1/col
# ----- Desktop -----

# ----- Laptop -----
# ! python -m pip install -r "/home/jared/Desktop/Fall2025/CS-6353/Homewo
# ----- Laptop -----
```

```

/home/jared/CS-6353/Homeworks/assignment1/cs6353/datasets
--2025-09-05 13:03:14-- http://www.cs.toronto.edu/~kriz/cifar-10-python.t
ar.gz
Resolving www.cs.toronto.edu (www.cs.toronto.edu)... 128.100.3.30
Connecting to www.cs.toronto.edu (www.cs.toronto.edu)|128.100.3.30|:80...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 170498071 (163M) [application/x-gzip]
Saving to: 'cifar-10-python.tar.gz'

```

```

cifar-10-python.tar 100%[=====>] 162.60M 18.4MB/s in 9.0
s

```

```

2025-09-05 13:03:23 (18.0 MB/s) - 'cifar-10-python.tar.gz' saved [17049807
1/170498071]

```

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cifar-10-batches-py/
cifar-10-batches-py/data_batch_4
cifar-10-batches-py/readme.html
cifar-10-batches-py/test_batch
cifar-10-batches-py/data_batch_3
cifar-10-batches-py/batches.meta
cifar-10-batches-py/data_batch_2
cifar-10-batches-py/data_batch_5
cifar-10-batches-py/data_batch_1
/home/jared/CS-6353/Homeworks/assignment1
Requirement already satisfied: anyio==3.7.1 in /home/jared/anaconda3/envs/
CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homework
s/assignment1/colab_requirements.txt (line 1)) (3.7.1)
Requirement already satisfied: appnope==0.1.3 in /home/jared/anaconda3/env
s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homewor
ks/assignment1/colab_requirements.txt (line 2)) (0.1.3)
Requirement already satisfied: argon2-cffi==23.1.0 in /home/jared/anaconda
3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Ho
meworks/assignment1/colab_requirements.txt (line 3)) (23.1.0)
Requirement already satisfied: argon2-cffi-bindings==21.2.0 in /home/jare
d/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/
CS-6353/Homeworks/assignment1/colab_requirements.txt (line 4)) (21.2.0)
Requirement already satisfied: arrow==1.2.3 in /home/jared/anaconda3/envs/
CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homework
s/assignment1/colab_requirements.txt (line 5)) (1.2.3)
Requirement already satisfied: asttokens==2.2.1 in /home/jared/anaconda3/e
nvs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homew
orks/assignment1/colab_requirements.txt (line 6)) (2.2.1)
Requirement already satisfied: async-lru==2.0.4 in /home/jared/anaconda3/e
nvs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homew
orks/assignment1/colab_requirements.txt (line 7)) (2.0.4)
Requirement already satisfied: attrs==23.1.0 in /home/jared/anaconda3/env
s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homewor
ks/assignment1/colab_requirements.txt (line 8)) (23.1.0)
Requirement already satisfied: Babel==2.12.1 in /home/jared/anaconda3/env
s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homewor
ks/assignment1/colab_requirements.txt (line 9)) (2.12.1)
Requirement already satisfied: backcall==0.2.0 in /home/jared/anaconda3/en
vs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homewo
rks/assignment1/colab_requirements.txt (line 10)) (0.2.0)
Requirement already satisfied: beautifulsoup4==4.12.2 in /home/jared/anaco
nda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-635
3/Homeworks/assignment1/colab_requirements.txt (line 11)) (4.12.2)
Requirement already satisfied: bleach==6.0.0 in /home/jared/anaconda3/env

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s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 12)) (6.0.0)
Requirement already satisfied: certifi==2023.7.22 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 13)) (2023.7.22)
Requirement already satisfied: cffi==1.15.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 14)) (1.15.1)
Requirement already satisfied: charset-normalizer==3.2.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 15)) (3.2.0)
Requirement already satisfied: comm==0.1.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 16)) (0.1.4)
Requirement already satisfied: contourpy==1.1.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 17)) (1.1.0)
Requirement already satisfied: cycycler==0.11.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 18)) (0.11.0)
Requirement already satisfied: debugpy==1.6.7.post1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 19)) (1.6.7.post1)
Requirement already satisfied: decorator<=5.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 20)) (4.4.2)
Requirement already satisfied: defusedxml==0.7.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 21)) (0.7.1)
Requirement already satisfied: executing==1.2.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 22)) (1.2.0)
Requirement already satisfied: fastjsonschema==2.18.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 23)) (2.18.0)
Requirement already satisfied: fonttools==4.42.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 24)) (4.42.1)
Requirement already satisfied: fqdn==1.5.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 25)) (1.5.1)
Requirement already satisfied: idna==3.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 26)) (3.4)
Requirement already satisfied: imageio==2.31.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 27)) (2.31.1)
Requirement already satisfied: ipykernel<=5.5.6 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 28)) (5.5.6)
Requirement already satisfied: ipython<=7.34.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 29)) (7.34.0)
Requirement already satisfied: isoduration==20.11.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 30)) (20.11.0)
Requirement already satisfied: jedi==0.19.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 31)) (0.19.0)
Requirement already satisfied: Jinja2==3.1.2 in /home/jared/anaconda3/env

s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 32)) (3.1.2)
Requirement already satisfied: json5==0.9.14 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 33)) (0.9.14)
Requirement already satisfied: jsonpointer==2.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 34)) (2.4)
Requirement already satisfied: jsonschema==4.19.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 35)) (4.19.0)
Requirement already satisfied: jsonschema-specifications==2023.7.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 36)) (2023.7.1)
Requirement already satisfied: jupyter-events==0.7.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 37)) (0.7.0)
Requirement already satisfied: jupyter-lsp==2.2.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 38)) (2.2.0)
Requirement already satisfied: jupyter_client<8.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 39)) (7.4.9)
Requirement already satisfied: jupyter_core==5.3.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 40)) (5.3.1)
Requirement already satisfied: jupyter_server==2.7.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 41)) (2.7.2)
Requirement already satisfied: jupyter_server_terminals==0.4.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 42)) (0.4.4)
Requirement already satisfied: jupyterlab==4.0.5 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 43)) (4.0.5)
Requirement already satisfied: jupyterlab-pygments==0.2.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 44)) (0.2.2)
Requirement already satisfied: jupyterlab_server==2.24.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 45)) (2.24.0)
Requirement already satisfied: kiwisolver==1.4.5 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 46)) (1.4.5)
Requirement already satisfied: MarkupSafe==2.1.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 47)) (2.1.3)
Requirement already satisfied: matplotlib==3.7.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 48)) (3.7.2)
Requirement already satisfied: matplotlib-inline==0.1.6 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 49)) (0.1.6)
Requirement already satisfied: mistune==3.0.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 50)) (3.0.1)
Requirement already satisfied: nbclient==0.8.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 51)) (0.8.0)

Requirement already satisfied: nbconvert==7.7.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 52)) (7.7.4)

Requirement already satisfied: nbformat==5.9.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 53)) (5.9.2)

Requirement already satisfied: nest-asyncio==1.5.7 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 54)) (1.5.7)

Requirement already satisfied: notebook_shim==0.2.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 55)) (0.2.3)

Requirement already satisfied: numpy<1.24,>=1.22 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 56)) (1.23.5)

Requirement already satisfied: overrides==7.4.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 57)) (7.4.0)

Requirement already satisfied: packaging==23.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 58)) (23.1)

Requirement already satisfied: pandas<=1.5.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 59)) (1.5.3)

Requirement already satisfied: pandocfilters==1.5.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 60)) (1.5.0)

Requirement already satisfied: parso==0.8.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 61)) (0.8.3)

Requirement already satisfied: pexpect==4.8.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 62)) (4.8.0)

Requirement already satisfied: pickleshare==0.7.5 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 63)) (0.7.5)

Requirement already satisfied: Pillow==10.0.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 64)) (10.0.0)

Requirement already satisfied: platformdirs==3.10.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 65)) (3.10.0)

Requirement already satisfied: prometheus-client==0.17.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 66)) (0.17.1)

Requirement already satisfied: prompt-toolkit==3.0.39 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 67)) (3.0.39)

Requirement already satisfied: psutil==5.9.5 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 68)) (5.9.5)

Requirement already satisfied: ptyprocess==0.7.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 69)) (0.7.0)

Requirement already satisfied: pure-eval==0.2.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 70)) (0.2.2)

Requirement already satisfied: pycparser==2.21 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 71)) (2.21)

Requirement already satisfied: Pygments==2.16.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 72)) (2.16.1)

Requirement already satisfied: pyparsing==3.0.9 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 73)) (3.0.9)

Requirement already satisfied: python-dateutil==2.8.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 74)) (2.8.2)

Requirement already satisfied: python-json-logger==2.0.7 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 75)) (2.0.7)

Requirement already satisfied: pytz==2023.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 76)) (2023.3)

Requirement already satisfied: PyYAML==6.0.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 77)) (6.0.1)

Requirement already satisfied: pyzmq<25 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 78)) (24.0.1)

Requirement already satisfied: referencing==0.30.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 79)) (0.30.2)

Requirement already satisfied: requests==2.31.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 80)) (2.31.0)

Requirement already satisfied: rfc3339-validator==0.1.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 81)) (0.1.4)

Requirement already satisfied: rfc3986-validator==0.1.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 82)) (0.1.1)

Requirement already satisfied: rpds-py==0.9.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 83)) (0.9.2)

Requirement already satisfied: scipy==1.11.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 84)) (1.11.2)

Requirement already satisfied: seaborn==0.12.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 85)) (0.12.2)

Requirement already satisfied: Send2Trash==1.8.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 86)) (1.8.2)

Requirement already satisfied: six==1.16.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 87)) (1.16.0)

Requirement already satisfied: sniffio==1.3.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 88)) (1.3.0)

Requirement already satisfied: soupsieve==2.4.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 89)) (2.4.1)

Requirement already satisfied: stack-data==0.6.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 90)) (0.6.2)

Requirement already satisfied: terminado==0.17.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 91)) (0.17.1)

Requirement already satisfied: tinycss2==1.2.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 92)) (1.2.1)

Requirement already satisfied: tornado<=6.3.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 93)) (6.3.2)

Requirement already satisfied: traitlets==5.9.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 94)) (5.9.0)

Requirement already satisfied: tzdata==2023.3 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 95)) (2023.3)

Requirement already satisfied: uri-template==1.3.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 96)) (1.3.0)

Requirement already satisfied: urllib3==2.0.4 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 97)) (2.0.4)

Requirement already satisfied: wcwidth==0.2.6 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 98)) (0.2.6)

Requirement already satisfied: webcolors==1.13 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 99)) (1.13)

Requirement already satisfied: webencodings==0.5.1 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 100)) (0.5.1)

Requirement already satisfied: websocket-client==1.6.2 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 101)) (1.6.2)

Requirement already satisfied: exceptiongroup in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from anyio==3.7.1->-r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 1)) (1.3.0)

Requirement already satisfied: typing-extensions>=4.0.0 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from async-lru==2.0.4->-r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 7)) (4.15.0)

Requirement already satisfied: tomli in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from jupyterlab==4.0.5->-r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 43)) (2.2.1)

Requirement already satisfied: ipython-genutils in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from ipykernel<=5.5.6->-r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 28)) (0.2.0)

Requirement already satisfied: setuptools>=18.5 in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from ipython<=7.34.0->-r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 29)) (78.1.1)

Requirement already satisfied: entrypoints in /home/jared/anaconda3/envs/CS6353/lib/python3.10/site-packages (from jupyter_client<8.0->-r /home/jared/CS-6353/Homeworks/assignment1/colab_requirements.txt (line 39)) (0.4)

k-Nearest Neighbor (kNN) exercise

Complete and hand in this completed worksheet (including its outputs and any supporting code outside of the worksheet) with your assignment submission. For more details see the [assignments page](#) on the course website.

The kNN classifier consists of two stages:

- During training, the classifier takes the training data and simply remembers it
- During testing, kNN classifies every test image by comparing to all training images and transferring the labels of the k most similar training examples
- The value of k is cross-validated

In this exercise you will implement these steps and understand the basic Image Classification pipeline, cross-validation, and gain proficiency in writing efficient, vectorized code.

```
In [2]: # Run some setup code for this notebook.
from __future__ import print_function

import random
import numpy as np
from cs6353.data_utils import load_CIFAR10
import matplotlib.pyplot as plt

# This is a bit of magic to make matplotlib figures appear inline in the
# rather than in a new window.
%matplotlib inline
plt.rcParams['figure.figsize'] = (10.0, 8.0) # set default size of plots
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'

# Some more magic so that the notebook will reload external python module
# see http://stackoverflow.com/questions/1907993/autoreload-of-modules-in
%load_ext autoreload
%autoreload 2
```

```
In [3]: # Load the raw CIFAR-10 data.
# ----- Laptop -----
# cifar10_dir = '/home/jared/Desktop/Fall2025/CS-6353/Homeworks/assignmen
# ----- Laptop -----

# ----- Desktop -----
cifar10_dir = '/home/jared/CS-6353/Homeworks/assignment1/cifar-10-batches
# ----- Desktop -----

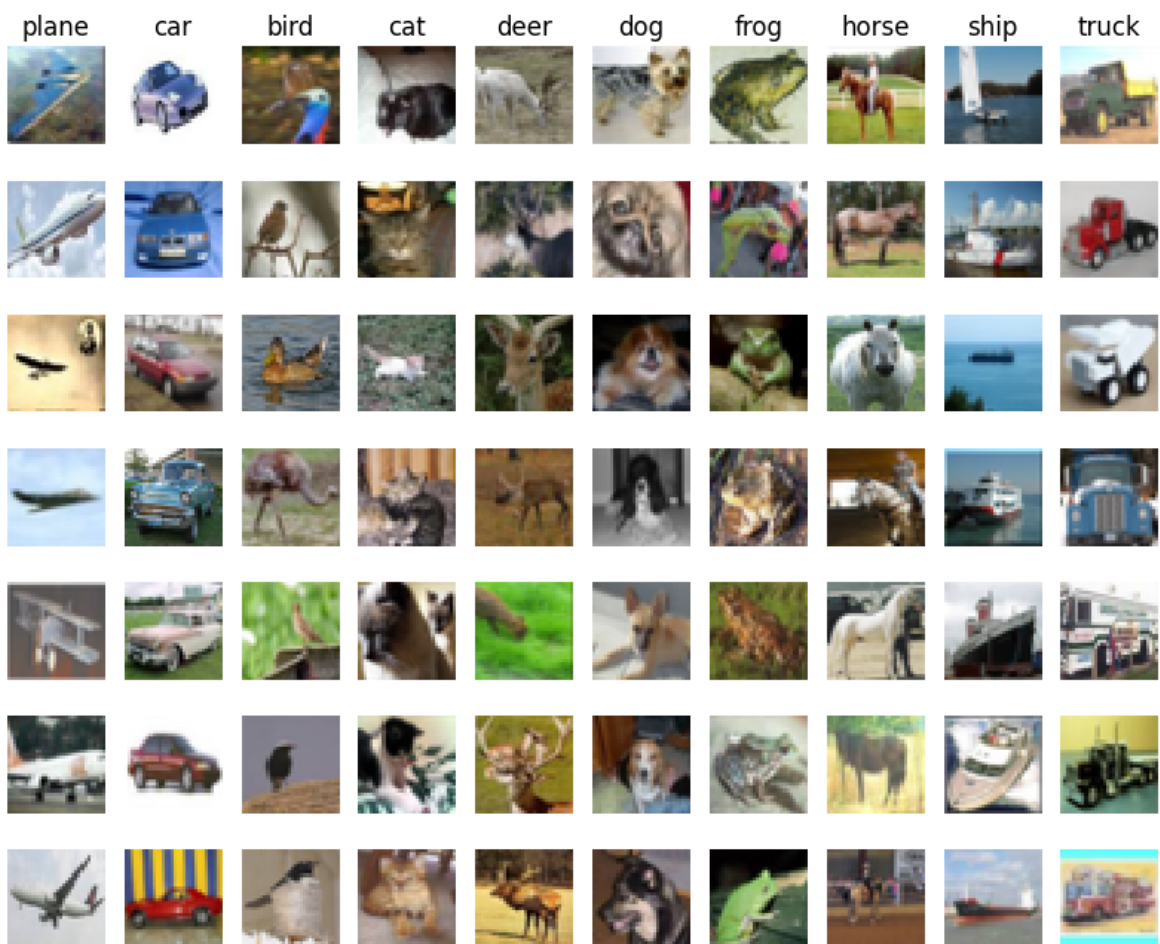
# Cleaning up variables to prevent loading data multiple times (which may
try:
    del X_train, y_train
    del X_test, y_test
    print('Clear previously loaded data.')
except:
    pass

X_train, y_train, X_test, y_test = load_CIFAR10(cifar10_dir)

# As a sanity check, we print out the size of the training and test data.
print('Training data shape: ', X_train.shape)
print('Training labels shape: ', y_train.shape)
print('Test data shape: ', X_test.shape)
print('Test labels shape: ', y_test.shape)
```

Training data shape: (50000, 32, 32, 3)
 Training labels shape: (50000,)
 Test data shape: (10000, 32, 32, 3)
 Test labels shape: (10000,)

```
In [4]: # Visualize some examples from the dataset.
# We show a few examples of training images from each class.
classes = ['plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck']
num_classes = len(classes)
samples_per_class = 7
for y, cls in enumerate(classes):
    idxs = np.flatnonzero(y_train == y)
    idxs = np.random.choice(idxs, samples_per_class, replace=False)
    for i, idx in enumerate(idxs):
        plt_idx = i * num_classes + y + 1
        plt.subplot(samples_per_class, num_classes, plt_idx)
        plt.imshow(X_train[idx].astype('uint8'))
        plt.axis('off')
        if i == 0:
            plt.title(cls)
plt.show()
```



```
In [5]: # Subsample the data for more efficient code execution in this exercise
num_training = 5000
mask = list(range(num_training))
X_train = X_train[mask]
y_train = y_train[mask]

num_test = 500
mask = list(range(num_test))
X_test = X_test[mask]
```

```
y_test = y_test[mask]
```

```
In [6]: # Reshape the image data into rows
X_train = np.reshape(X_train, (X_train.shape[0], -1))
X_test = np.reshape(X_test, (X_test.shape[0], -1))
print(X_train.shape, X_test.shape)

(5000, 3072) (500, 3072)
```

```
In [7]: from cs6353.classifiers import KNearestNeighbor

# Create a kNN classifier instance.
# Remember that training a kNN classifier is a noop:
# the Classifier simply remembers the data and does no further processing
classifier = KNearestNeighbor()
classifier.train(X_train, y_train)
```

We would now like to classify the test data with the kNN classifier. Recall that we can break down this process into two steps:

1. First we must compute the distances between all test examples and all train examples.
2. Given these distances, for each test example we find the k nearest examples and have them vote for the label

Lets begin with computing the distance matrix between all training and test examples. For example, if there are **N_{tr}** training examples and **N_{te}** test examples, this stage should result in a **N_{te} x N_{tr}** matrix where each element (i,j) is the distance between the i -th test and j -th train example.

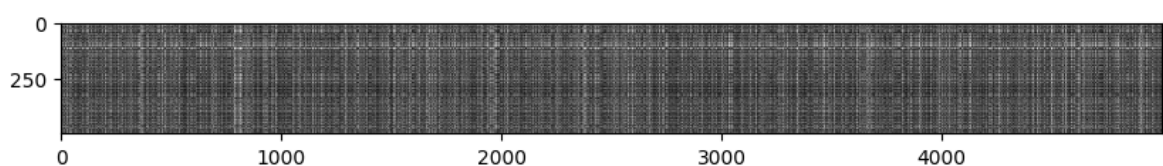
First, open `cs6353/classifiers/k_nearest_neighbor.py` and implement the function `compute_distances_two_loops` that uses a (very inefficient) double loop over all pairs of (test, train) examples and computes the distance matrix one element at a time.

```
In [8]: # Open cs6353/classifiers/k_nearest_neighbor.py and implement
# compute_distances_two_loops.

# Test your implementation:
dists = classifier.compute_distances_two_loops(X_test)
print(dists.shape)

(500, 5000)
```

```
In [9]: # We can visualize the distance matrix: each row is a single test example
# its distances to training examples
plt.imshow(dists, interpolation='none')
plt.show()
```



Inline Question #1: Notice the structured patterns in the distance matrix, where some

rows or columns are visible brighter. (Note that with the default color scheme black indicates low distances while white indicates high distances.)

- What in the data is the cause behind the distinctly bright rows?
- What causes the columns?

Your Answer:

- If we have a full row of distinctly bright lines that means that the training example was "far" away from everything in our training set. This means that we have an image that our model thinks is unique.
- If we have a full column of distinctly bright lines that means that the specific training example associated with that column is "far" away from every single test example.

```
In [10]: # Now implement the function predict_labels and run the code below:
# We use k = 1 (which is Nearest Neighbor).
y_test_pred = classifier.predict_labels(dists, k=1)

# Compute and print the fraction of correctly predicted examples
num_correct = np.sum(y_test_pred == y_test)
accuracy = float(num_correct) / num_test
print('Got %d / %d correct => accuracy: %f' % (num_correct, num_test, acc
```

Got 137 / 500 correct => accuracy: 0.274000

You should expect to see approximately 27% accuracy. Now let's try out a larger k , say $k = 5$:

```
In [11]: y_test_pred = classifier.predict_labels(dists, k=5)
num_correct = np.sum(y_test_pred == y_test)
accuracy = float(num_correct) / num_test
print('Got %d / %d correct => accuracy: %f' % (num_correct, num_test, acc
```

Got 139 / 500 correct => accuracy: 0.278000

You should expect to see a slightly better performance than with $k = 1$.

Inline Question 2 We can also choose other distance metrics such as L1 distance. The performance of a Nearest Neighbor classifier that uses L1 distance will not change if (Select all that apply.):

1. The data is preprocessed by subtracting the mean.
2. The data is preprocessed by subtracting the mean and dividing by the standard deviation.
3. The coordinate axes for the data are rotated.
4. None of the above. (Mean and standard deviation in (1) and (2) are vectors and can be different across dimensions)

Your Answer: (1), (2) {See Image Below For Explanation}

Your explanation:

$$X_1^{(1)} = [x_1^{(1)}, x_2^{(1)}, \dots, x_D^{(1)}]$$

$$L1(x_1, x_2) = \sum_{i=1}^D |x_i^{(1)} - x_i^{(2)}|$$

$$X_2^{(2)} = [x_1^{(2)}, x_2^{(2)}, \dots, x_D^{(2)}]$$

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X^{(i)}$$

$$L1(x_1 - \bar{X}, x_2 - \bar{X}) = \sum |x_i^{(1)} - \bar{X} - (x_i^{(2)} - \bar{X})|$$

$$= \sum |x_i^{(1)} - \cancel{\bar{X}} - x_i^{(2)} + \cancel{\bar{X}}|$$

$$L1(x_1 - \bar{X}, x_2 - \bar{X}) = \sum |x_i^{(1)} - x_i^{(2)}| = \sum_{i=1}^D |x_i^{(1)} - x_i^{(2)}|$$

∴ When we subtract the mean the distance remains the same so performance is unaffected

$$\sum_{i=1}^D \left| \frac{x_i^{(1)}}{\sigma} - \frac{x_i^{(2)}}{\sigma} \right|$$

$$\sum_{i=1}^D \left| \frac{1}{\sigma} (x_i^{(1)} - x_i^{(2)}) \right|$$

Same distance scaled by a constant. All terms are scaled by this constant so performance is unaffected when dividing by σ

In [12]: # Now lets speed up distance matrix computation by using partial vectoriz
with one loop. Implement the function compute_distances_one_loop and ru
code below:

```
dists_one = classifier.compute_distances_one_loop(X_test)
```

```
# To ensure that our vectorized implementation is correct, we make sure t  
# agrees with the naive implementation. There are many ways to decide whe  
# two matrices are similar; one of the simplest is the Frobenius norm. In  
# you haven't seen it before, the Frobenius norm of two matrices is the s  
# root of the squared sum of differences of all elements; in other words,  
# the matrices into vectors and compute the Euclidean distance between th  
difference = np.linalg.norm(dists - dists_one, ord='fro')  
print('Difference was: %f' % (difference, ))  
if difference < 0.001:  
    print('Good! The distance matrices are the same')  
else:  
    print('Uh-oh! The distance matrices are different')
```

Difference was: 0.000000

Good! The distance matrices are the same

In [13]: # Now implement the fully vectorized version inside compute_distances_no_
and run the code

```
dists_two = classifier.compute_distances_no_loops(X_test)
```

```
# check that the distance matrix agrees with the one we computed before:  
difference = np.linalg.norm(dists - dists_two, ord='fro')
```

```

print('Difference was: %f' % (difference, ))
if difference < 0.001:
    print('Good! The distance matrices are the same')
else:
    print('Uh-oh! The distance matrices are different')

```

Difference was: 0.000000

Good! The distance matrices are the same

```

In [14]: # Let's compare how fast the implementations are
def time_function(f, *args):
    """
    Call a function f with args and return the time (in seconds) that it
    """
    import time
    tic = time.time()
    f(*args)
    toc = time.time()
    return toc - tic

two_loop_time = time_function(classifier.compute_distances_two_loops, X_t
print('Two loop version took %f seconds' % two_loop_time)

one_loop_time = time_function(classifier.compute_distances_one_loop, X_te
print('One loop version took %f seconds' % one_loop_time)

no_loop_time = time_function(classifier.compute_distances_no_loops, X_tes
print('No loop version took %f seconds' % no_loop_time)

# you should see significantly faster performance with the fully vectoriz

```

Two loop version took 23.845083 seconds

One loop version took 21.769915 seconds

No loop version took 0.173854 seconds

Cross-validation

We have implemented the k-Nearest Neighbor classifier but we set the value $k = 5$ arbitrarily. We will now determine the best value of this hyperparameter with cross-validation.

```

In [15]: num_folds = 5
k_choices = [1, 3, 5, 8, 10, 12, 15, 20, 50, 100]

X_train_folds = []
y_train_folds = []
#####
# TODO:
# Split up the training data into folds. After splitting, X_train_folds a
# y_train_folds should each be lists of length num_folds, where
# y_train_folds[i] is the label vector for the points in X_train_folds[i]
# Hint: Look up the numpy array_split function.
#####
# Your code
X_train_folds = np.array_split(X_train, num_folds)
y_train_folds = np.array_split(y_train, num_folds)

#####
#
END OF YOUR CODE

```

```
#####

# A dictionary holding the accuracies for different values of k that we f
# when running cross-validation. After running cross-validation,
# k_to_accuracies[k] should be a list of length num_folds giving the diff
# accuracy values that we found when using that value of k.
k_to_accuracies = {}

#####
# TODO:
# Perform k-fold cross validation to find the best value of k. For each
# possible value of k, run the k-nearest-neighbor algorithm num_folds tim
# where in each case you use all but one of the folds as training data an
# last fold as a validation set. Store the accuracies for all fold and al
# values of k in the k_to_accuracies dictionary.
#####
# Your code
for cur_k in k_choices:
    for i in range(num_folds):
        # Pull the validation folds out
        cur_x_validation = X_train_folds[i]
        cur_y_validation = y_train_folds[i]

        # Concat all folds besides the validation folds
        cur_x_train = np.concatenate([X_train_folds[j] for j in range(num
        cur_y_train = np.concatenate([y_train_folds[j] for j in range(num

        # Append all but the current fold together
        classifier.train(cur_x_train, cur_y_train)

        # Get accuracy with the current fold
        y_test_pred = classifier.predict(cur_x_validation, k=cur_k)
        num_correct = np.sum(y_test_pred == cur_y_validation)
        accuracy = float(num_correct) / num_test
        if cur_k in k_to_accuracies:
            k_to_accuracies[cur_k].append(accuracy)
        else:
            # must be initialized to a list for append to work later
            k_to_accuracies[cur_k] = [accuracy]

#####
#                                     END OF YOUR CODE
#####

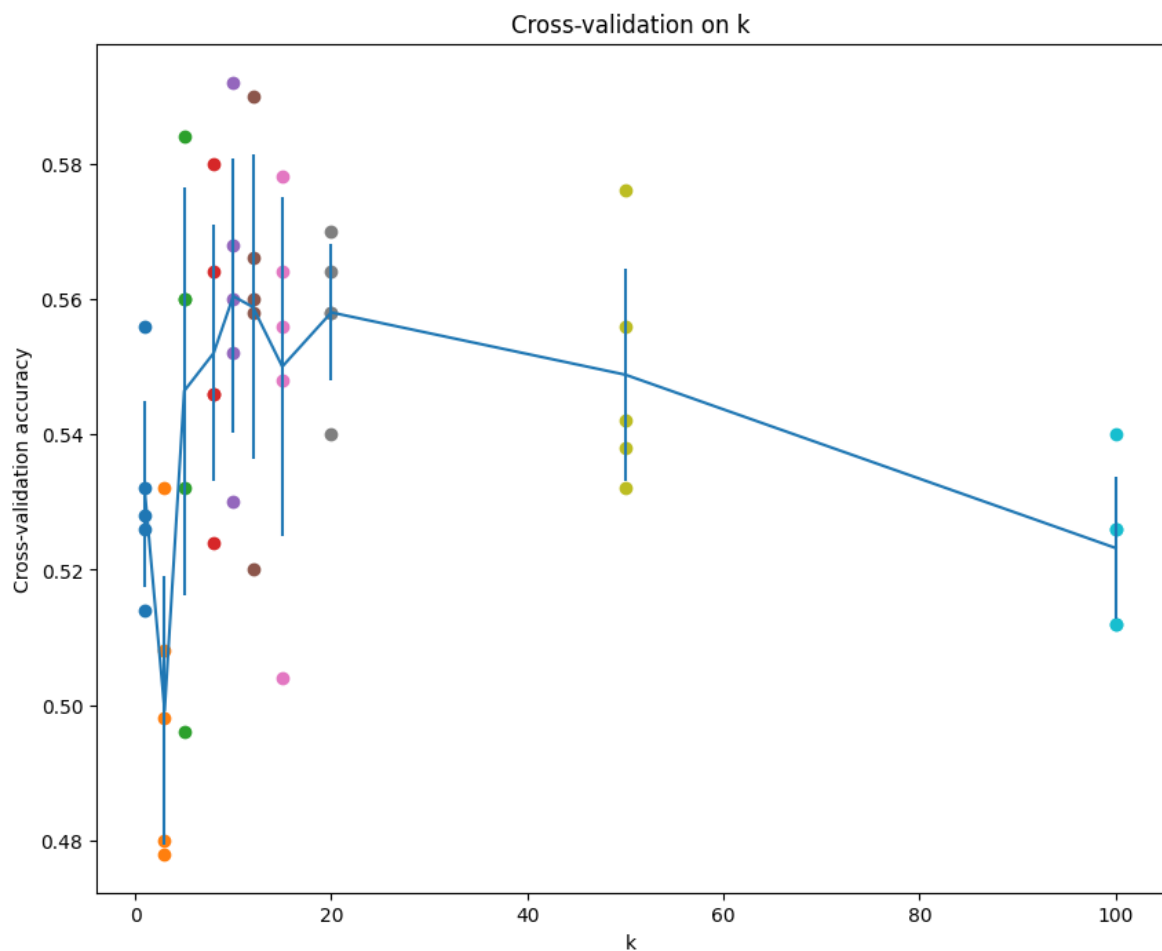
# Print out the computed accuracies
for k in sorted(k_to_accuracies):
    for accuracy in k_to_accuracies[k]:
        print('k = %d, accuracy = %f' % (k, accuracy))
```

```
k = 1, accuracy = 0.526000
k = 1, accuracy = 0.514000
k = 1, accuracy = 0.528000
k = 1, accuracy = 0.556000
k = 1, accuracy = 0.532000
k = 3, accuracy = 0.478000
k = 3, accuracy = 0.498000
k = 3, accuracy = 0.480000
k = 3, accuracy = 0.532000
k = 3, accuracy = 0.508000
k = 5, accuracy = 0.496000
k = 5, accuracy = 0.532000
k = 5, accuracy = 0.560000
k = 5, accuracy = 0.584000
k = 5, accuracy = 0.560000
k = 8, accuracy = 0.524000
k = 8, accuracy = 0.564000
k = 8, accuracy = 0.546000
k = 8, accuracy = 0.580000
k = 8, accuracy = 0.546000
k = 10, accuracy = 0.530000
k = 10, accuracy = 0.592000
k = 10, accuracy = 0.552000
k = 10, accuracy = 0.568000
k = 10, accuracy = 0.560000
k = 12, accuracy = 0.520000
k = 12, accuracy = 0.590000
k = 12, accuracy = 0.558000
k = 12, accuracy = 0.566000
k = 12, accuracy = 0.560000
k = 15, accuracy = 0.504000
k = 15, accuracy = 0.578000
k = 15, accuracy = 0.556000
k = 15, accuracy = 0.564000
k = 15, accuracy = 0.548000
k = 20, accuracy = 0.540000
k = 20, accuracy = 0.558000
k = 20, accuracy = 0.558000
k = 20, accuracy = 0.564000
k = 20, accuracy = 0.570000
k = 50, accuracy = 0.542000
k = 50, accuracy = 0.576000
k = 50, accuracy = 0.556000
k = 50, accuracy = 0.538000
k = 50, accuracy = 0.532000
k = 100, accuracy = 0.512000
k = 100, accuracy = 0.540000
k = 100, accuracy = 0.526000
k = 100, accuracy = 0.512000
k = 100, accuracy = 0.526000
```

```
In [16]: # plot the raw observations
         for k in k_choices:
             accuracies = k_to_accuracies[k]
             plt.scatter([k] * len(accuracies), accuracies)

         # plot the trend line with error bars that correspond to standard deviation
         accuracies_mean = np.array([np.mean(v) for k,v in sorted(k_to_accuracies.items())])
         accuracies_std = np.array([np.std(v) for k,v in sorted(k_to_accuracies.items())])
         plt.errorbar(k_choices, accuracies_mean, yerr=accuracies_std)
```

```
plt.title('Cross-validation on k')
plt.xlabel('k')
plt.ylabel('Cross-validation accuracy')
plt.show()
```



```
In [17]: # Based on the cross-validation results above, choose the best value for
# retrain the classifier using all the training data, and test it on the
# data. You should be able to get above 28% accuracy on the test data.
best_k = 10

classifier = KNearestNeighbor()
classifier.train(X_train, y_train)
y_test_pred = classifier.predict(X_test, k=best_k)

# Compute and display the accuracy
num_correct = np.sum(y_test_pred == y_test)
accuracy = float(num_correct) / num_test
print('Got %d / %d correct => accuracy: %f' % (num_correct, num_test, acc
```

Got 141 / 500 correct => accuracy: 0.282000

Inline Question 3 Which of the following statements about k -Nearest Neighbor (k -NN) are true in a classification setting, and for all k ? Select all that apply.

1. The training accuracy of a 1-NN (nearest neighbor) will always be better than or equal to that of 5-NN.
2. The test accuracy of a 1-NN will always be better than that of a 5-NN.
3. The decision boundary of the k -NN classifier is linear.
4. The time needed to classify a test example with the k -NN classifier grows with the

size of the training set.

5. It remembers all of the training data.
6. remembers all of the test data.
7. KNN cannot be applied for distance metrics other than L1 and L2.
8. KNN does more computation during test time than training time.
9. KNN does more computation during training time than test time.
10. KNN does equal computation during training time and test time.
11. KNN does equal computation during training time and test time.

Your Answer: 1, 4, 5, 8

Your explanation:

- (1) This is because we store images exactly so if we have a 1-NN each training example will directly match up with the training example it was matched on.
- (4) This is because every time we want to classify a test example we have to compare that example to every single other example in our training set
- (5) All a KNN does during training time is store all training examples
- (8) Training time just stores all training examples. Test time has to compare the test example to every single training example